

Digital Twin 2.0: Smart Logistics Infrastructure Enabling Agentic AI and Physical AI

Preview

Digital Twin (Digital Twin) has been one of the most highlighted digital technologies in logistics and supply-chain management over the past ten years, yet despite large-scale investment, it has remained limited to modest success, failing to deliver results beyond 3-D visualization systems.

The rapidly advancing large-language-model (LLM)-based Agentic AI and Physical AI used in robotics present a new breakthrough for logistics decision-making, but actual logistics sites are complex and training data for AI development are absolutely scarce. Digital twin is being redefined as the infrastructure of the AI era and is being elevated to Digital Twin 2.0. First, it serves as a virtual learning space that generates data resembling reality as synthetic training data for AI; and it functions as a validation space where AI safety and generalization capabilities can be evaluated before AI decisions are applied to reality.

This white paper examines three misconceptions surrounding digital twins: that they are a 100 % visual replica of reality; that they extend simulation; and that real-time bidirectional synchronization is essential. It emphasizes that the essence of a digital twin lies in constructing an environment that supports optimal AI decision-making in complex business settings.





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01.

Digital Twin, Beyond Misconception to Essence

Digital Twin, Beyond Misconception to Essence

Digital Twin has been one of the most prominent digital technologies over the past decade. There are various definitions of the concept of digital twin, but in the study by Semeraro et al. (2021) the concepts defined in various papers and research were summarized as follows.¹

"A set of adaptive models that emulate the behavior of a physical system in a virtual system getting real-time data to update itself along its life cycle. The digital twin replicates the physical system to predict failures and opportunities for changing, to prescribe real-time actions for optimizing and/or mitigating unexpected events observing and evaluating the operating profile system."

Interestingly, the conceptual roots of digital twin go back to the 1960s. The simulator that NASA used to model the Apollo space missions was its prototype; when the Apollo 13 oxygen-tank explosion occurred in 1970, NASA used a ground-based replica simulator of Apollo 13 to virtually recreate the spacecraft situation and find a way to return the crew.² Later, David Gelernter's 1991 book *Mirror Worlds* laid the ideological foundation, and in 2002 Dr. Michael Grieves of the University of Michigan academically formalized it, while the term "digital twin" itself is reported to have been first used in 2010 by NASA engineer John Vickers.³ This trajectory shows that digital twin is not a fleeting fad but a concept that has been refined over decades, and that it is now poised for another leap together with advances in AI technology.

Looking at recent digital-twin cases, Rolls-Royce applied a digital twin to a system that collects data from sensors installed on aircraft engines and analyzes it in real time, achieving predictive maintenance and fuel-efficiency improvements; Siemens built a metaverse-based digital twin for a smart factory and reported productivity gains of more than 20%.⁴ These results demonstrate that a digital twin can recreate actual industrial environments and contribute to increased productivity.

Digital Twin, Limited Success

As expectations for digital twins grew, global research institutes and consulting firms projected rapid annual growth in the related market, and pilot digital twin projects were launched across various industries such as manufacturing, logistics, and construction. Moreover, governments worldwide made large-scale investments, recognizing digital twins as a core infrastructure for digital transformation.

However, despite these massive investments, successful cases have been limited, and digital twin projects have failed to deliver tangible results beyond 3D visualizations that mimic reality. While demonstrations have shown warehouses and factories displayed in real time on large monitors, few examples exist of commercialized decision-support systems that effectively manage actual manufacturing and logistics operations. Consequently, before discussing new possibilities for digital twins, it is necessary to clarify the misconceptions held by the industry about digital twins.

1 Semeraro, Concetta, et al. "Digital twin paradigm: A systematic literature review." *Computers in industry* 130 (2021): 103469.

2 Allen, B. Danette. "Digital twins and living models at NASA." *Digital Twin Summit*. 2021.

3 https://en.wikipedia.org/wiki/Digital_twin

4 <https://ismguide.com/ism-digital-twin-case-studies/>

Three Misconceptions about Digital Twins

Misconception 1.

A digital twin is a visualization system that shows reality 100 % as it is

The most common misunderstanding is to think of a digital twin as a 3D graphic that simultaneously displays a logistics warehouse and a factory. In case studies where digital-twin systems have been implemented, flashy 3D visualizations of industrial sites often stand out, but this is merely the outward appearance of a successful digital-twin system—not its purpose. The core is to predict problems, analyze them, and provide real-time response measures within the twin itself.

Many failed digital-twin projects invest heavily in the visualization layer while the purpose—what they actually want to achieve with the twin—is unclear, or the decision-making process for finding algorithms and optimal responses is limited from a technical standpoint. Even in the McKinsey report, examples of successful digital twins stress the need to build them for real decision-making rather than for 3D-graphic-based visualization.⁵ For supply-chain decisions, if you collect the necessary portions of historical performance data and build a system that operates on that basis, it can be a digital twin even without any 3D graphics.

Misconception 2. A digital Twin is an extension of simulation

The second misconception is to regard a digital twin as a more advanced form of a traditional simulation system. If one looks only at the visualization aspect, digital twins and simulations share similarities. However, traditional simulation is a one-off system built to focus on specific domains and functions in order to achieve a particular purpose—such as productivity analysis for new equipment investment decisions or analysis of operating policies. Once the purpose of the simulation is fulfilled through scenario analysis, the simulation model is no longer needed. In other words, simulation can be described as a limited model created for a specific objective—such as investment analysis or policy analysis—rather than a virtual world tied to actual reality.

In contrast, a digital twin should aim to create a digital model of the real world and, based on that, continuously develop a model that assists in decision-making. Instead of repeatedly building, analyzing, and discarding a model for each specific purpose, a process is required that continuously reflects changes in reality in the model so that it can be incorporated into the decision-making process. If simulation is used as part of ongoing decision-making, such a simulation model could be called a digital twin.

Misconception 3. A digital twin must be 100 % linked to reality in real time

When we review existing research on digital twins, there is a tendency to emphasize real-time synchronization. Data from sensors installed in the physical world are uploaded to the virtual model in real time, and the optimal decisions made in the virtual model are fed back to reality; this two-way synchronization is presented as an essential condition for a digital twin.

5 Mickael Brossard et al., "Digital twins: When and why to use one," McKinsey & Company, April 30, 2024

This definition can be justified when a digital twin is used as a system-monitoring or real-time control tool. To manage warehouses, ports, power plants, etc., in real time, a virtual model that is 100 % identical to the actual site—achieved through real-time synchronization—can be helpful.

However, even in such cases, a digital twin does not necessarily need to be 100% linked to reality in real time. The purpose of a digital twin is to provide the data needed to operate the physical world optimally and to verify the results of real-time decision making. Depending on its purpose, a lower level of data linkage may be sufficient.

Beyond Misconception to Essence: The Rediscovery of Digital Twins Is No Accident

When we sort out the three misconceptions, the core purpose that digital twins aim for becomes clear. A digital twin is not valuable for creating a virtual model of the real world; rather, it is needed as a tool to support optimal decision-making in a constantly changing, complex business environment. The goal is to validate various scenarios against real data, and based on that, to optimize response measures in real time and apply them to reality.

Viewing digital twins in this way makes it clear why they have re-emerged after a period of stagnation. A technology has appeared that can supply the data required for logistics and supply-chain decision-making and deliver optimal response strategies effectively in real time. The rise of AI—rapidly advancing since the launch of ChatGPT in 2023—has highlighted the need for digital twins. The emergence of a novel decision-making tool is redefining why digital twins are necessary.

The LLM AI technology—trained on text, images, and video from the Internet—is advancing beyond merely understanding natural language and generating documents to a stage where it can perform complex reasoning and decision-making. These models are called foundation models because the same AI can be applied to a wide range of purposes rather than a single, specific task. As AI has reached this level, attempts to embed it in real-world business have accelerated. Representative outcomes include Agentic AI—AI agents that communicate with users and proactively carry out tasks using an LLM-based foundation model as the software’s brain—and Physical AI, which embeds AI as the “brain” of robots or autonomous transport vehicles. In both cases, the brain that drives software or hardware is no longer a custom algorithm crafted for a particular situation but a universally developed LLM AI, opening up numerous possibilities.

However, when we try to apply AI to real-world settings, a decisive limitation emerges: where to train these AIs and where to validate their decisions. As foundation models are built for general purposes, they are likely to have limited ability to handle actual field data in logistics and supply chains. Consequently, the need to retrain them using on-site data becomes increasingly important.

If we collect logistics and supply-chain data, train on it, and deploy AI in the decision-making process, the industrial environment tolerates no trial-and-error, and the training data required for decision optimization are woefully insufficient. Even when we aim to support demand forecasting and inventory optimization, performance data exist only for a limited historical period; once conditions change, all that data becomes unusable. The same applies to robot learning for logistics automation—if the products being handled change, new training must be conducted.

01. Digital Twin, Beyond Misconception to Essence

This is precisely where the strength of a digital twin appears. If a digital system that mirrors the real site exists, we can obtain enough data to train AI. Moreover, before AI makes decisions in the actual field, its safety and generalizability can be verified within the digital system. A digital twin is not valuable merely for visualizing and synchronizing real-time data for monitoring; it serves as a training ground for AI intended for logistics sites and as a test environment to validate AI decisions. In this context, 3-D visualization and real-time synchronization become ancillary functions; the crucial role is to support AI training and validation.

Ultimately, the redefinition of digital twins that is gaining momentum—"Digital Twin 2.0"—can be seen as a natural extension generated by the success of LLM-based AI. Traditional digital twins had limitations because human intervention or static, human-crafted algorithms were embedded in the decision-making process, reducing effectiveness in reality. Now that AI powered by LLMs is ready for full-scale deployment in industry, Digital Twin 2.0, combined with Agentic AI and Physical AI, establishes a foundation for explosive growth.

[Table 1] Comparison of Digital Twin 1.0 and 2.0

Category	Digital Twin 1.0 (Traditional)	Digital Twin 2.0 (This Whitepaper)
Core Purpose	Visual replication and monitoring of reality	AI-driven autonomous operation and optimization
Key Functions	3D visualization, real-time status display	AI learning and validation infrastructure
Visualization Emphasis	Essential (what you see is core)	Optional (used when needed)
Simulation Relationship	Operated as a separate tool (used for human decision-making)	Real-time simulation embedded (focused on AI learning and validation)
Reality Synchronization	Emphasis on real-time sync	Synchronization sufficient for AI learning and validation
Data Utilization	Collect all available on-site data and use for human decision-making	Collect data within the scope needed for AI learning and validation
Decision-making Entity	Human (dashboard-based interpretation)	AI agent (autonomous judgment, execution)
Operational Mode	Operate based on predefined algorithms	Agentic AI-based proactive operation
Relationship with AI	Indirectly linked to AI (can provide data)	Integration with Agentic AI and Physical AI is mandatory
Success Criteria	Visualization quality and scope of real-time data sync	Establishment of autonomous operation sites based on AI decision-making

(Source: Author)

02.

Overcome Challenges of Logistics Decision-Making and Limits of AI Learning

Complexity of Logistics Systems and New Opportunities with AI

Logistics systems are built as highly complex structures in which numerous systems are interconnected. Rather than being simple mechanical equipment or processes, they consist of a massive network in which warehouses and transportation services are linked like a web. Demand does not move as predicted, and supply can be abruptly disrupted. As automation systems within warehouses become more advanced, workers and automated equipment become entangled and do not operate as intended. Complex decisions such as forecasting demand, replenishing inventory, or assigning orders to logistics equipment must be made in real time. This volatility and complexity are inherent characteristics of the logistics industry.

Traditionally, logistics decision-making has involved two approaches: a manual, experience- and intuition-driven method that relies on on-site experts, and a tool-based method built on logical algorithms such as mathematical optimization and data analysis. On-site experts can make decisions suited to the field based on deep understanding and know-how, but as problems become more complex they have struggled to derive optimal solutions quickly. Conversely, algorithmic decision-making offers objective and accurate optimal solutions for complex problems, but it often relies on fragmented and oversimplified data about the field, limiting its applicability in practice.

In a complex logistics system, both intuition-based approaches and algorithm-driven intelligent systems reduce field usability. It is precisely at this point that attempts are being made to harness AI for a breakthrough. When trained on sufficient data, AI can rapidly generate responses that suit the field and has a structural characteristic that allows various types of data to be input.

Virtual Data for AI Training Created by Digital Twin

When properly managed, a digital twin can generate massive amounts of data that AI can learn from. It can also create virtual scenarios that have never occurred in reality. By freely generating various scenarios—such as port closures, sudden demand drops, freight-rate spikes, natural disasters, infrastructure disruptions, large-scale returns—within a virtual environment and feeding them to AI, the AI can learn optimal response strategies for a wide range of situations.⁶

One could also use actual performance data to forecast demand or optimize inventory, and deploy robots on the shop floor to sort and package products, thereby generating data and training AI. However, data obtained this way tends to be limited, concentrating on certain scenarios. In contrast, a digital twin has the strength of generating learning data from events that have never happened before, allowing AI to be trained on optimal response measures. By creating data in a virtual environment and repeatedly training on it, it is possible to simultaneously secure the advantage of devising optimal responses based on field experts' know-how and logical algorithms.

Another noteworthy point is that a digital twin can simulate the outcomes of decisions. When AI proposes a decision appropriate to the current situation, it is difficult to be certain that applying it directly in the real world will produce the predicted result. In such cases, a digital twin can be used to quickly predict the ripple effects on logistics and supply chains caused by that decision. The ability to generate virtual supply-chain events and forecast their future impacts—something hard to verify in reality—is a unique strength of the digital twin. Data that can teach how certain actions produce specific outcomes when problems arise is difficult to obtain in the real world. The direction that digital twins should pursue lies precisely here.

⁶ Aloysius Yap, "From visibility to foresight: The technology bolstering supply chain resilience," DHL Logistics of Things

03.

Digital Twin: a Learning and a Validation Space of AI



AI Learning Through Trial and Error

The 2016 Go match between AlphaGo and 9-dan player Lee Se-dol marked a major turning point in AI history. However, AlphaGo's innovation lay not in simply memorizing past game records as correct answers, but in learning by playing countless simulated games within a Digital Twin that enables virtual matches and discovering on its own which moves are good. Historical Go game data provide valuable performance data for AI training, but they do not help the system learn a variety of scenarios. This learning method, in which AI tries actions directly and improves by observing the outcomes, is called reinforcement learning. Reinforcement learning has achieved great success in games and robot control, and it is now being applied in earnest to logistics and supply-chain operations.⁷

For reinforcement learning to work, three things must be supplied to the AI. First, information about the current situation is needed—for example, how much inventory is on hand, the backlog of orders, the locations of robots, and where vehicles are moving. Second, a list of candidate actions that the AI can choose from must be provided, such as how much to order, how to move robots, or which route to use for shipping cargo. Third, a reward for the outcome of each action is required. The AI must be told how much cost the chosen action incurs, whether productivity increases, whether transport time decreases, and so on, so it can evaluate performance. By endlessly presenting and evaluating diverse scenarios, the AI learns decision-making strategies suitable for many situations encountered in logistics and supply chains. Without a digital twin, this would be infeasible.

⁷ Rolf, Benjamin, et al. "A review on reinforcement learning algorithms and applications in supply chain management." *International Journal of Production Research* 61.20 (2023): 7151-7179.

Testing Decisions Before Real-World Implementation

Even when AI is trained with ample data, applying its decisions directly to reality can be risky. Logistics systems may encounter problems, and robots could create hazardous situations for operators. A digital twin can offer a space where AI decisions are verified across various dimensions before being deployed in the real world.

In typical operating conditions, test scenarios can be used to compare AI decisions with traditional methods to assess effectiveness. Moreover, predefined scenarios can introduce events such as order fluctuations, system interruptions, or equipment failures, allowing evaluation of the AI's decisions. It is also possible to assess whether the AI contributes not only to efficiency but also to the safe operation of logistics and supply chains. For example, one can verify that robots prioritize worker safety even if productivity drops, or examine whether inventory optimization leads to the bullwhip effect. This is similar with how autonomous vehicles undergo massive, impossible-in-reality virtual driving tests before hitting actual roads.

An even more critical aspect is confirming that the AI performs well in novel situations. AI trained solely on real-world data often yields unexpected results when faced with unfamiliar scenarios. It may handle known environments adeptly but behave unpredictably in untrained ones. A digital twin can generate scenarios that differ completely from past conditions, providing a foundation for evaluating the AI's "generalization" capabilities. Thus, a digital twin serves not only as a space to train and develop AI but also as a verification arena to assess whether the AI's capabilities are sufficient for real-world deployment.

Positioning digital twins as a component of AI-driven smart logistics and smart SCM decision-making processes helps explain why earlier digital-twin implementations fell short and clarifies the direction digital twins should pursue moving forward.

04.

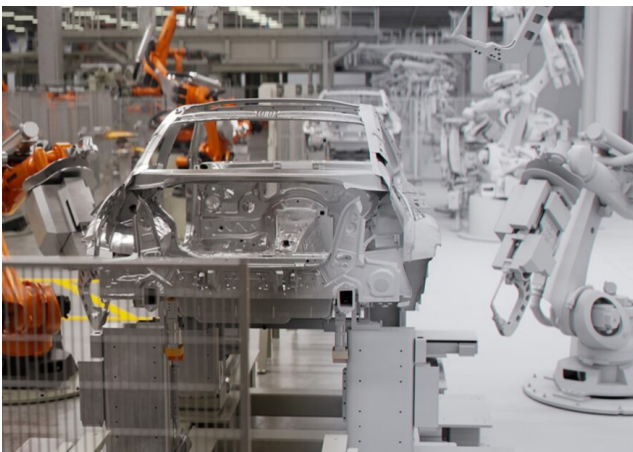
Cases of Digital Twin Utilization by Global Companies

BMW and NVIDIA: Training Factory Robots in a Virtual Environment

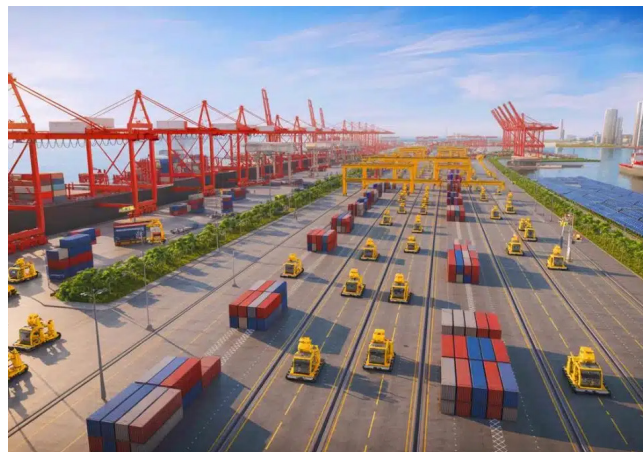
Germany's BMW has built a digital twin of its real automobile factory on NVIDIA's digital twin platform, Omniverse, and trained AI to operate automated equipment—such as autonomous mobile robots (AMRs)—within the virtual car factory.⁸ The BMW virtual plant realized through the digital twin is not merely a 3D visualization model; it replicates the workings of the physical world—dynamics, friction, lighting, obstacles, worker pathways—creating a physics based environment that is almost identical to reality. As the AI learns a variety of scenarios for operating the factory within the digital twin, it comes to understand BMW's plant, establishing an environment that can be deployed in the real world. Based on these results, BMW announced plans to implement a future manufacturing plant run by AI. According to NVIDIA, Pepsi has created a digital twin of its logistics warehouse on the Omniverse platform, and FedEx's digital twin for training logistics robots is also operating on the Omniverse platform.

Singapore Tuas Container Port: Simultaneous Control of Hundreds of Autonomous Vehicles

Regarded as the world's largest automated port, Singapore's PSA Tuas Port has built a digital twin to optimize container port operations and has tested a system that uses AI to operate logistics automation equipment such as hundreds of automated guided vehicles (AGVs).⁹ Creating a mathematical optimization algorithm that allows AGVs to move autonomously without colliding in a complex container terminal is difficult and impractical, and it's impossible for human operators to manually control each vehicle. The introduction of AI trained on abundant data from the digital twin is seen as a breakthrough. As container ports become highly automated, AI adoption has become a prerequisite. Not only in Singapore's Tuas Port, but also in world class ports such as the Rotterdam Port in the Netherlands, efforts are underway to implement digital twin technology for AI learning in logistics environments.



BMW's Digital Twin case⁸



Singapore's Tuas Port Digital Twin case⁹

⁸ NVIDIA Korea, "BMW group's innovation continuing with NVIDIA Omniverse" NVIDIA Blog Korea"

⁹ 'They're Moving 65 Million Without Humans': World's Largest Automated Port Set to Rewrite the Rules of Global Shipping Forever," Sustainability Times, 2025,

05.

Digital Twin that Completes Smart Logistics/SCM

Digital Twin that Completes Smart Logistics/SCM

As we have seen so far, digital twins are not merely impressive 3D visual graphics; they are creating new value as spaces that generate the data needed for AI training and verify AI decision-making. However, no matter how well an AI is trained within a digital twin, if the digital twin does not match reality, the AI cannot be deployed in the real world. This discrepancy is called the sim-to-real gap, and it will be the key to successful digital-twin adoption. It is impossible for the virtual world realized by a digital twin to encompass all environmental variables that arise in real-world logistics and supply chains. Digital twins are built by focusing on the parts needed for AI training, which inevitably leaves out many real-world constraints.

Therefore, a digital twin should not be treated as a one-time implementation that works out of the box; it must be operated as a continuously upgraded system after an initial prototype is built. After training and validating AI with virtual data and scenarios, any problems that arise when applying it to reality should be reflected back into the digital twin to adjust the environment, thereby linking the physical and virtual worlds and requiring an advanced, iterative evolution. This is where a digital-twin construction project differentiates itself from conventional smart-logistics projects. Viewed positively, companies that adopt digital twins early can use the trial-and-error process to secure deployable Agentic AI and Physical AI, giving them a competitive advantage.

What Should Be Done

As we have seen, when logistics companies invest in digital twins, they should not focus on “how realistic it looks” but rather evaluate “how well they can train the AI needed for real-world operations.” Instead of emphasizing flashy 3-D graphics, the focus must be on whether the digital twin provides an environment where AI can be trained and validated. Implementing a digital twin should be regarded not as a simple IT expense but as a strategic investment to secure an AI-based competitive advantage.

The logistics industry has long dreamed of digital transformation and the realization of smart logistics. Digital twins—especially those designed for AI training and validation—are expected to become the catalyst that turns this dream into reality. As AI-agent technology and Physical-AI technology are reshaping the paradigm of smart logistics, investment in digital twins became a task that can’t be postponed anymore. I hope logistics companies will actively pursue AI training and validation through digital twins, using it as an opportunity to lift their global logistics competitiveness to the next level.



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